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An Explainable Machine Learning Model for Water Potability Assessment Using SHAP and Hybrid Feature Selection

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***Abstract*—** One of the most important public health needs is access to safe drinking water, and preventing waterborne illnesses requires determining the potability of the water. Through the analysis of several physicochemical factors, predictive modeling provides an effective method for assessing the quality of water. In this work, a comprehensive dataset for classifying potable and non-potable water was created by combining three publicly accessible water quality datasets that were gathered from Kaggle. The suggested methodology addresses class imbalance and ensures data quality by integrating feature scaling, class balancing using SMOTE, and median-based missing value imputation. Key features like hardness, pH, and turbidity were identified by feature selection utilizing Mutual Information (MI) and Extra Trees Classifier (ETC), whereas less significant features like conductivity and organic carbon were eliminated. Cross-validation was used to assess several machine learning models and ensemble techniques in order to measure performance with reliability. SHAP analysis was used to make the model interpretable and revealed that the most important parameters influencing predictions were hardness, pH, and trihalomethanes. Following five-fold cross-validation, the Extra Trees Classifier demonstrated the efficacy and dependability of the suggested preprocessing, feature selection, and modeling technique for water potability prediction with an Accuracy of 95.4%, Precision of 94.83%, and F1-score of 95.43%.

Keywords*— *Potable and non-potable water, feature scaling, feature selection, interpretable machine learning, data balancing

I. Introduction

Access to safe and clean drinking water is essential for both public health and human life. A significant cause of waterborne illnesses is still contaminated water, particularly in areas without efficient water quality monitoring infrastructure [1]. Even if they are accurate, traditional water testing methods require specialized laboratory facilities and are time and resource expensive [2]. As machine learning and water quality datasets become more accessible, prediction algorithms have become a viable substitute for quickly determining the potability of water based on physicochemical factors [3].

Although machine learning has been widely investigated for water quality prediction, several limitations still exist. The model's capacity to be applied to a variety of water conditions is limited by the fact that many research only use one dataset. Predictions that are unstable are frequently the result of improper handling of missing values, class imbalance, and irrelevant features [4]. Furthermore, interpretability is usually disregarded, depriving decision-makers of knowledge about the factors influencing the categorization outcomes. Furthermore, confidence in stated performance is diminished in the absence of thorough model comparisons and rigorous cross-validation.

Public health depends on having access to safe and drinkable water, yet in many areas, water quality monitoring is still constrained by reliance on laboratories, expensive testing, and slow reaction times. While machine learning for water potability prediction has been investigated in a number of publications, the majority are limited by single-dataset reliance, unbalanced data, inadequate preprocessing, and interpretability issues. These restrictions limit the model's ability to generalize and make it less applicable in the actual world. Furthermore, decision-makers are unable to comprehend which water-quality characteristics influence categorization results due to the lack of explainable frameworks. The suggested framework performs best when utilizing ETC, showing excellent predictive power and transparency, which makes it a useful tool for aiding in policy decisions and water quality management.

The point below is the contribution of proposed work.

- Following schema alignment and duplicate reduction, three publicly accessible Kaggle datasets were combined to create a heterogeneous dataset with 6,842 distinct samples that represented a variety of water sources.
- Reliable model training was ensured by using SMOTE to fix class imbalance and median imputation to handle missing variables.
- To improve feature robustness and interpretability, a combined MI+ ETC ranking was devised to concurrently capture statistical relevance and model-based importance.
- To evaluate model predictions and identify the physicochemical factors most significantly affecting potability classification, SHAP analysis was utilized.

II. Related Work

El-Bacha Rachid et al. [5] used Support Vector Machine and Random Forest on physicochemical data, the study used machine learning to evaluate the potability of the water. With a ROC–AUC of 0.75, 70% accuracy, 72% precision, and

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improved performance, RF demonstrated promise for real-time monitoring. Modest predictive performance, dependence on a single dataset, lack of preprocessing and tuning information, explainability analysis, and lack of real-world validation are some of the drawbacks.

Jinal Patel et al. [6] used Z-score normalization and SMOTE for balance, the study compared SVM, Decision Tree, Random Forest, Gradient Boost, and AdaBoost on the Kaggle Water Quality Index dataset in order to address diminishing water quality. The best accuracy of 81% was attained by RF and GB. Explainable AI (XAI) with LIME was used to find important features in order to increase model transparency. However, the work's poor accuracy, dependence on a single dataset, and lack of wider validation for practical use are its limitations. N. Laya et al. [7] used machine learning models, such as RF, KNN, SVM, Decision Tree, and LR, along with a 10-feature dataset to estimate the potability of the water. The maximum accuracy was attained by RF after parameter adjustment, increasing from 80% to 81%. In order to efficiently classify potable and non-potable water and facilitate improved water quality management, the finished model was preserved with the pickle library for use in future forecasts. Anurag Mukati et al. [8] employed RF, DTC and XGBoost models to forecast drinking water safety using the Water Potability dataset, which consists of 3,276 samples. Following review and preprocessing, Random Forest had the best accuracy, at 70%. The results back up better public health protection, policymaking, and water safety procedures. Ivan Ivanov [9] showed Water potability is critical in ecology because it affects habitable areas, health, and food quality. With dwindling drinkable water supplies, water quality prediction has become a critical environmental issue. This study introduces a simple, improved machine learning technique for forecasting water potability that is more easily applied and adaptable to different datasets. The method allows for rapid evaluation of new water sources, which aids environmental monitoring and public health operations. Robert G. De Luna et al. [10] used Univariate Selection, Recursive Feature Elimination, and Feature Importance, this study employs machine learning to predict the potability of the water and pinpoint important features. With an accuracy of 89%, the ETC outperformed the other classifiers tested, providing important information for managing water resources and public health.

Diwash Poudel et al. [11] used a statistically imputed dataset, this study forecasts the potability of water by substituting medians for missing pH, chloramine, and trihalomethane values. After testing four algorithms (LR, K-NN, RF, and ANN), Random Forest (700 trees, depth 30) showed the best results. To enhance forecasts, further optimization is required. Partha Sarathi Bishnu et al. [12] used actual datasets, this chapter investigates the application of machine learning (ML) and deep learning (DL) approaches to forecast the potability of water. In order to improve water quality prediction, a number of models were evaluated and their outcomes described. These models included gradient boosting, random forest, decision tree, logistic regression, k-NN, multilayer perceptron, additional trees, and naive Bayes.

III. Methodology

In Fig. 1 the suggested paradigm for predicting the potability of water follows a systematic and repeatable sequential sequence. To enhance generalization capabilities and boost data diversity, three publicly accessible Kaggle datasets were first gathered, schema-aligned, and merged after duplicate records were eliminated. During preprocessing, all numerical features were normalized by feature scaling to provide uniform contribution during model training, while missing values and possible outliers in physicochemical characteristics were handled using median imputation, which is robust to skewed distributions. This allowed for an objective assessment of the model by splitting the dataset into groups of 80% training and 20% testing. By using SMOTE, bias toward the dominant class was lessened and the class imbalance between potable and non-potable samples was lessened. Seven features were then chosen using a hybrid feature selection approach that combined MI and ETC importance to account for both statistical dependency and model-based relevance. Twelve machine learning models were then trained, including SVM, RF, BNB, ETC, LR, AB, LDA, KNN, QDA, GNB, DTC, and GB, as well as soft voting ensemble combinations. To lower the risk of overfitting, 5-fold cross-validation was used to further evaluate the model's resilience. Accuracy, precision, recall, F1-score, and ROC-AUC measures were used to thoroughly assess performance. In order to evaluate feature contributions and improve the transparency and usefulness of the suggested approach for water potability assessment, SHAP analysis was finally carried out.

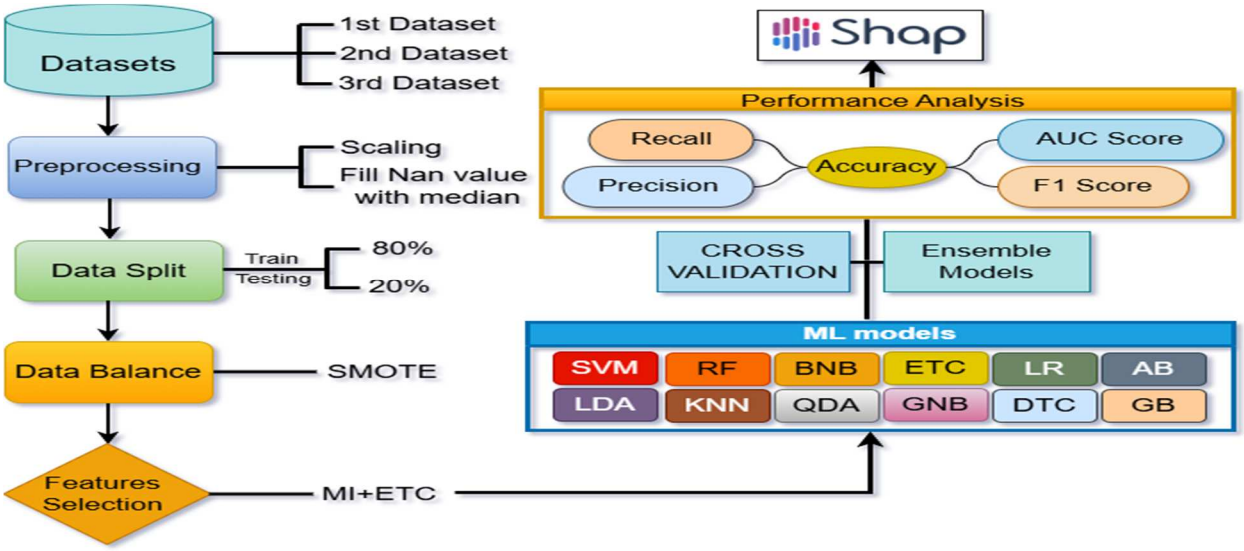


Fig. 1. Workflow of proposed model

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A. Dataset

The first dataset [13], which had 3,276 samples with some missing values, the second [14], which contained 1,272 samples without any missing values, and the third [15], which contained 2,294 samples without any missing values, were all gathered from Kaggle. In order to verify that the feature names, data types, and units of the three Kaggle datasets were the same, a schema alignment check was initially performed. Following verification that every dataset had the same ten physicochemical parameters, the datasets were combined row-wise to create a single dataset. All feature columns were used to identify duplicate data, which were then eliminated before analysis to prevent redundancy. Extreme or inconsistent values were regarded as outliers and substituted with the median values for each attribute. The final combined dataset had 6,842 distinct samples, increasing the predictive model's generality and variety across geographical sources.

B. Feature Selection

TABLE I provides a summary of the feature selection procedure based on the combined ranks and selection status of MI scores and ETC importance. Different components of feature relevance are captured by MI and ETC significance. While ETC significance represents model-based contribution within an ensemble of randomized trees, MI measures the non-linear relationship between each feature and the objective variable without assuming any particular model. Strong statistical association (MI) and high predictive contribution (ETC) traits are simultaneously emphasized when both scores are combined. The risk of choosing traits that seem significant solely under one metric is decreased by this hybrid ranking. Hardness was a top-selected feature along with pH (rank 5), Turbidity, Trihalomethanes, Solids, Chloramines (all rank 10), and Sulfate (rank 11). Its MI score was 0.158809, its ETC significance was 0.114024, and its combined rank was 4. With the lowest significance scores of 16 and 14, respectively, conductivity and organic_carbon was excluded from the final model. This selection procedure eliminates less significant features while emphasizing the parameters that have the greatest influence on the predicting task. Select the eight features from 10 features according to the TABLE I.

TABLE I. FEATURE SELECTION BY MI AND ETC

Feature	MI Score	ETC Importance	Combined Rank	Status
Hardness	0.158809	0.114024	4	Selected
Turbidity	0.152968	0.101604	10	Selected
Trihalomethanes	0.15175	0.102537	10	Selected
ph	0.146855	0.127347	5	Selected
Solids	0.145091	0.111523	10	Selected
Chloramines	0.143638	0.111824	10	Selected
Conductivity	0.14198	0.100828	16	Not Selected
Organic_carbon	0.140362	0.103205	14	Not Selected
Sulfate	0.133548	0.127108	11	Selected

Figure 2 displays a bar chart with the group feature ranks for the various water quality parameters. A lower value indicates a higher level of importance. Hardness (rank 4), followed by pH (rank 5), is the most significant attribute. Trihalomethanes, solids, turbidity, and chloramines all get a rank of 10, suggesting minor importance, but sulfate has a slightly higher grade of 11. In contrast, the prediction model is least affected by organic carbon (rank 14) and conductivity (rank 16). All things considered, the analysis reveals that hardness and pH are the most crucial elements, whereas conductivity and organic carbon have minimal impact.

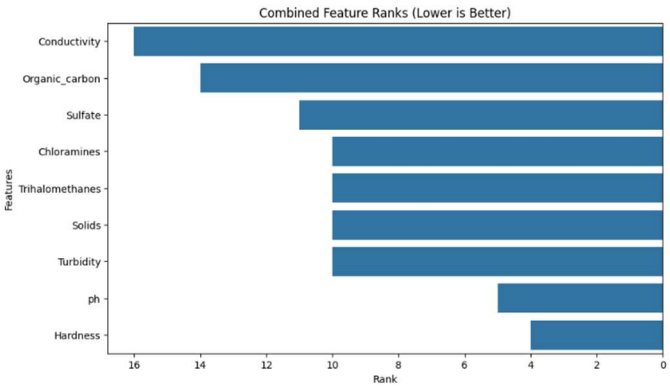


Fig. 2. Combined Feature Ranking using MI+ETC

C. Data Split

For accurate model evaluation, the dataset was split into 80% for training and 20% for testing [16]. By enabling efficient training of the model and providing an unbiased evaluation of its performance on anonymous data, this split yields robust and dependable findings.

D. Extra Tree Classifier

It generates several trees, each of which has a unique random subset of the training set's attributes and data. In contrast to traditional decision trees or random forests, the extra trees classifier introduces unpredictability by randomly selecting split points inside each feature. Better generalization and less variance result from increased unpredictability [17].

IV. RESULT AND DISCUSSION

Recall evaluates the ability to identify true positives, precision shows positive prediction accuracy, and accuracy gauges total correctness. While the ROC-AUC evaluates classification performance across a wide range of thresholds, the F1 Score strikes a compromise between precision and recall [18]. The suggested system facilitates dependable model training and manages the dataset effectively. The following equations were used to determine the performance parameter: (1), (2), (3), (4), and (5).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{3}$$

$$F_1 - Score = 2 * \left(\frac{Sensitivity * Precision}{Sensitivity + Precision}\right) \tag{4}$$

$$Auc = \sum_{i=1}^{n-1} (FPR_{i+1} - FPR_i).\left(\frac{TPR_{i+1} + TPR_i}{2}\right) \tag{5}$$

A. Confusion Metrics

In Fig. 3 Among individual models, the (a) ETC achieves the highest accuracy, with relatively few misclassifications and a good balance between genuine positives and true negatives, according to the confusion matrices. (c) LR and (b) RF both

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exhibit greater incorrect classifications but do very well. Due to an unbalanced error distribution, (d) RF+BNB performs poorly for voting ensembles, but (e) RF+ETC and (f) BNB+ETC both get good performance by utilizing complimentary strengths. Whether employed alone or in carefully selected ensembles, ETC provides the most precise, reliable, and well-rounded forecasts for the potability of water.

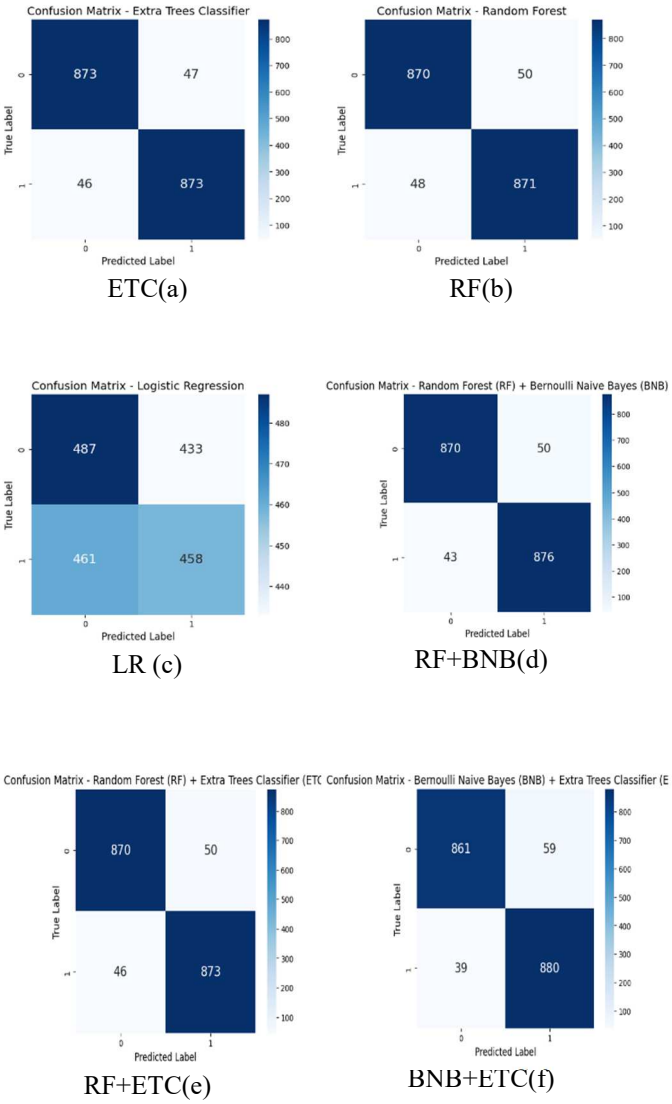


Fig. 3. Confusion metrics for ML and Ensemble models.

B. *Roc Curve*

In Fig. 4 the True Positive Rate versus False Positive Rate of the top five voting classifier combinations is displayed in the ROC curve. AUC values above 0.98 indicate great predictive performance, and all five models show excellent discriminative capacity. AUC of 0.9929 was the highest for the combination of BNB and ETC. RF with ETC (AUC = 0.9921) and ETC with AdaBoost (AUC = 0.9919) were the next closest. Even though RF with LR was the least successful of the top models, it managed to maintain a high AUC of 0.9880. According to the findings, ensemble approaches especially those that incorporate ETC and RF consistently produce better classification results for all tested combinations.

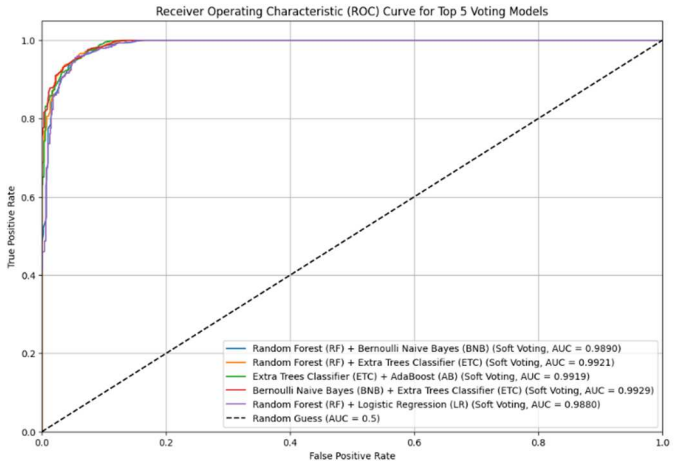


Fig. 4. Roc curve for top five ensemble models

C. *Figures and Tables*

In TABLE II the performance ratings of different machine learning models following feature selection are shown. RF came in second with an accuracy of 0.9467 and an AUC of 0.9889, while ETC had the best accuracy (0.9494) and AUC score (0.9929) of all the models. A great performance was also shown by the DTC, which had an accuracy of 0.9027. Strong recall (0.8705) and competitive accuracy (0.7912) were demonstrated by KNN. LR, LDA, and GNB had lower total scores, whereas BNB had perfect recall (1.0) but comparatively low accuracy (0.5008). Simpler classifiers were clearly outperformed by ensemble-based models (ETC, RF, and DTC), proving that feature selection combined with powerful ensemble approaches produces the best outcomes.

TABLE II. MACHINE LEARNING MODEL PARAMETERS SCORE WITH FEATURE SELECTION

Model	Accuracy	Precision	Recall	F1 Score	AUC Score
LR	0.513866	0.514029	0.498368	0.506077	0.516774
DTC	0.902664	0.918552	0.883569	0.900721	0.902654
SVM	0.686786	0.681481	0.700762	0.690987	0.766112
KNN	0.791191	0.751174	0.870511	0.806452	0.893546
GNB	0.569331	0.591631	0.446137	0.508685	0.610821
BNB	0.500816	0.500272	1	0.666909	0.50163
RF	0.94671	0.945711	0.947769	0.946739	0.988954
GB	0.69168	0.701835	0.665941	0.683417	0.783621
AdaBoost	0.578575	0.593506	0.49728	0.541149	0.610684
LDA	0.518216	0.518729	0.49728	0.507778	0.516459
QDA	0.628603	0.660326	0.528836	0.587311	0.696652
ETC	0.949429	0.948913	0.949946	0.949429	0.992943

In TABLE III displays, using soft voting following feature selection, the performance of twelve model combinations. The RF and BNB combination produced the highest accuracy (0.9494), a high AUC score (0.9889), and a strong recall (0.9532). ETC + AB and RF + LR both obtained 0.9473 accuracy, with RF + LR exhibiting the best precision (0.9507), while the RF + ETC pairing came in second with 0.9478 accuracy and a remarkable AUC of 0.9921. The combination of BNB + ETC has the highest AUC (0.9929) and an accuracy of 0.9467. Soft voting between complimentary models effectively improves predictive performance, as evidenced by the great performance of all combinations with little variance in accuracy.

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TABLE III. APPLY SOFT VOTING ON VARIOUS MODEL WITH FEATURE SELECTION

Models	Accuracy	Precision	Recall	F1 Score	AUC Score
(RF) + (BNB)	0.949429	0.946004	0.95321	0.949593	0.988954
(RF) + (ETC)	0.947798	0.945829	0.949946	0.947883	0.992106
(ETC) + (AB)	0.947254	0.944805	0.949946	0.947368	0.991929
(BNB) + (ETC)	0.94671	0.937167	0.957563	0.947255	0.992943
(RF) + (LR)	0.947254	0.950658	0.943417	0.947023	0.988043

In TABLE IV describes the outcomes of applying several machine learning models to 5-fold cross-validation. RF came in second with an accuracy of 0.9448 and an AUC of 0.9896, while ETC had the greatest accuracy (0.954) and AUC score (0.9931). DTC achieved an accuracy of 0.9046, demonstrating outstanding performance as well. Of the non-tree-based models, QDA received a score of 0.6332, while GB and SVM both had intermediate accuracies of 0.7047 and 0.6906, respectively. Lower performances were seen by models like BNB (0.4887), GNB (0.5520), AB (0.5711), and LR (0.5118). In general, ensemble and tree-based classifiers performed significantly better than more straightforward statistical models on every evaluation metric.

TABLE IV. PERFORMING 5-FOLD CROSS-VALIDATION

Model	Accuracy	Precision	Recall	F1 Score	AUC Score
ETC	0.954	0.9483	0.9605	0.9543	0.9931
RF	0.9448	0.949	0.9403	0.9446	0.9896
DTC	0.9046	0.9277	0.8776	0.9019	0.9047
GB	0.7047	0.709	0.6951	0.7019	0.7829
SVM	0.6906	0.6819	0.7147	0.6977	0.7706
QDA	0.6332	0.6609	0.5478	0.5989	0.703
AB	0.5711	0.5787	0.5347	0.5512	0.5995
LR	0.5118	0.5137	0.4935	0.5016	0.5176
GNB	0.552	0.5671	0.441	0.4955	0.5912
BNB	0.4887	0.3943	0.4002	0.262	0.5014

The accuracy of various machine learning models is examined in Fig. 5. Tree-based methods work well; of the models examined, ETC had the highest accuracy of 0.949, followed by RF at 0.947 and DTC at 0.903. While LR, GNB, BNB, and LDA fared badly, with accuracies ranging from 0.501 to 0.569, other models, including KNN and SVM, generated intermediate accuracy ratings of 0.791 and 0.687, respectively. Overall, the results show that ensemble-based approaches perform better than the other approaches in this comparison.

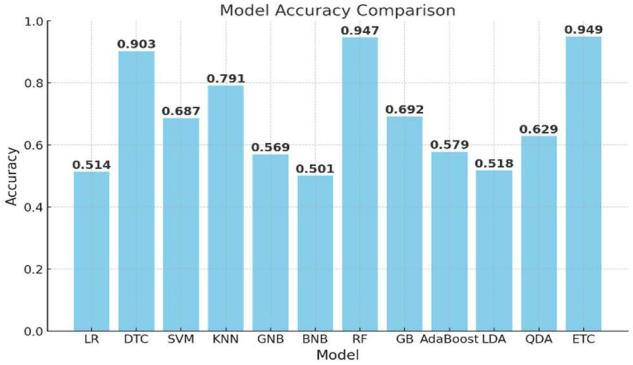


Fig. 5. Accuracy score of twelve models.

The top five model combinations in soft voting based on accuracy are displayed in Fig. 6. RF + BNB had the best accuracy at 0.9494, followed by RF + ETC at 0.9478. LR + ETC recorded 0.9467, while RF + LR and ETC + AB both scored 0.9473, indicating a little difference between the top performers. These results indicate that combining strong ensemble models usually yields very high accuracy. According to the smallest variation, some combinations are nearly equally useful for prediction.

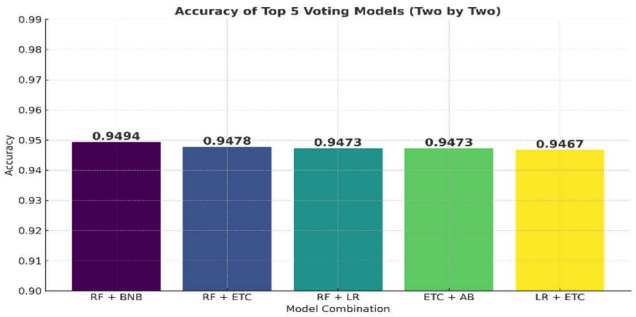


Fig. 6. Soft voting accuracy score of top five combination.

The performance of ten machine learning models is evaluated using 5-Fold Cross-Validation in Fig. 7. The bar chart using four important metrics: accuracy, precision, recall, and F1 Score. The Extra Trees Classifier (ETC) achieved the highest accuracy (95.4%), precision (94.83%), recall (96.05%), and F1 score (95.43%), outperforming all other models. Models like Bernoulli Naive Bayes (BNB) and Logistic Regression (LR) performed poorly, while Random Forest (RF) trailed closely behind. According to these results, the most effective and reliable model for this dataset is ETC.

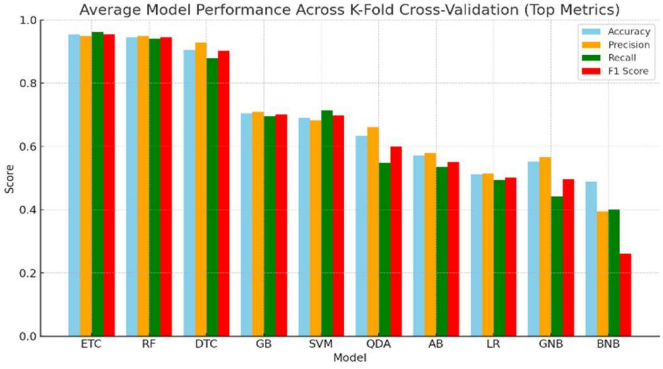


Fig. 7. 5-fold cross-validation of accuracy bar chart.

The Table V performance comparison offers convincing evidence of the superiority of our proposed work over previous studies. Our approach demonstrates a notable improvement over earlier research, even though previous works such as El-Bacha Rachid et al. [5] and Diwash Poudel et al. [11] only achieved 70% accuracy, and even Robert G. De Luna et al.'s best-performing earlier work [10] only managed 89% accuracy with the same dataset size (3,276 samples). In addition to surpassing all existing standards, we achieved a record-breaking accuracy of 95.4% by combining 6,842 sample records from combined datasets and using an enhanced Extra Trees Classifier (ETC) with 5-fold cross-validation. This shows the scalability, flexibility to larger datasets, and enhanced generalization ability of our approach. The significant improvement in accuracy demonstrates the robustness and practicality of our proposed method, making it the most effective water potability prediction system to date.

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TABLE V. PERFORMANCE ANALYSIS WITH OTHER WORK

Reference	Model	Dataset	Accuracy (%)
El-Bacha Rachid et al. [5]	SVM	2,011 sample records of dataset	70.00
Ivan Ivanov [9]	DTC	3,276 sample records of dataset	88.00
Robert G. De Luna [10]	ETC	3,276 sample records of dataset	89.00
Diwash Poudel [11]	RF	3,276 sample records of dataset	70.00
Proposed work	**Cross-validation (ETC)**	**6,842 sample records of dataset**	**95.4**

D. Interpretability

The SHAP summary for each feature and the average effect of the most important characteristics on the model's output value are shown in Fig. 8. Trihalomethanes, which significantly bias predictions in favor of the positive class, make the biggest positive contribution. Although to varying degrees, pH, hardness, and chloramines also have a beneficial effect. Among the characteristics displayed, turbidity has the least influence, whereas solids and sulfate have significant effects. The direction and degree of influence are represented by the blue-to-pink color gradient, where blue denotes negative effects and pink higher positive contributions. The model output value shift brought on by each characteristic is displayed on the horizontal axis.

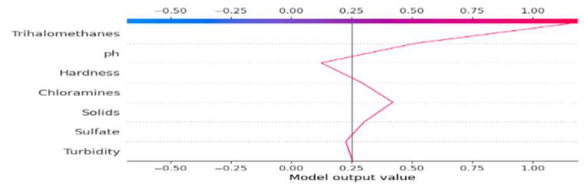


Fig. 8. SHAP feature importance for water potability prediction

In Fig. 9 This SHAP waterfall plot shows the relative contributions of each feature to the model's prediction, starting with the base value $E\ [f\ (X)]$ = 0.254 E[f(X)] = 0.254 and ending with the output $f\ (x)$ = 1.152 f(x)=1.152. Red indicates positive contributions, which raise the prediction. Trihalomethanes had the biggest positive impact (+0.64), followed by solids (+0.12), sulfate (+0.08), and pH (+0.39). The prediction is lowered by negative contributions (shown in blue), with the most significant impacts coming from Hardness (–0.17), Chloramines (–0.13), and Turbidity (–0.03). Overall, pH and trihalomethanes have a far stronger positive impact than a negative one, which raises the ultimate prediction.

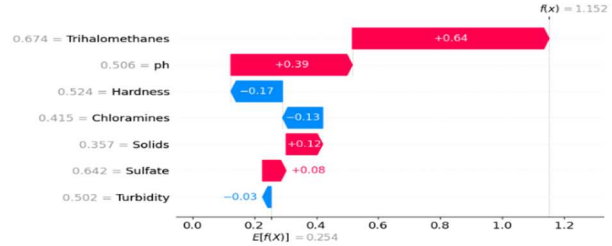


Fig. 9. SHAP waterfall plot for feature contribution in water potability prediction

V. CONCLUSION

This study combined three Kaggle datasets and used preprocessing, SMOTE-based balancing, and feature selection using MI and ETC to offer a strong machine learning framework for water potability prediction. Under 5-fold cross-validation, the Extra Trees Classifier performed the best, obtaining 95.4% accuracy, 94.83% precision, and 95.43% F1-score and interpretability based on SHAP. Reliance on attributes found in the source datasets and the lack of real-time data evaluation are among the limitations. These will be addressed in future research by adding more factors and testing on dynamic data for wider applicability.

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